

Detection of obfuscated Tor traffic based on bidirectional generative adversarial networks and vision transform

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ARTICLE INFO

Keywords:

BiGAN

Obfuscated detection

ViT

Tor traffic system

ABSTRACT

The Onion Router (TOR) network is a decentralized system of volunteer-run servers that aims to protect the anonymity and privacy of users by routing their internet traffic through a series of nodes. Individuals who use the TOR network may employ obfuscated traffic to conceal their internet activity from network administrators or security systems attempting to block or monitor them. Furthermore, some may use obfuscated Tor traffic to hide illegal activities, such as buying and selling illegal goods or accessing illegal services on the dark web. Despite efforts to identify and block Tor traffic, challenges remain, such as a limited set of features for identification, leading to false positives and negatives. To address these challenges, this paper proposes a novel approach using Visual Transformation (ViT), and augmentation by Bidirectional Generative Adversarial Networks (BiGAN). The proposed approach demonstrates superior performance on the ISCX-Tor2016 dataset, achieving 99.59% accuracy, 99.83% recall, 99.72% precision, and 99.78% F-score, thereby outperforming current state-of-the-art techniques.

1. Introduction

With the fast pace of Internet technology and the tremendous expansion of the World Wide Web (WWW), three types of networks have emerged: (i) Surface Web, which refers to websites that can be indexed by search engines such as Google, (ii) Deep Web, which includes websites that exist but are not indexed by search engines due to the vastness of the internet, such as medical information, and (iii) Dark Web, which requires specialized software to access the content of websites because it exists on darknets over the public internet, such as sites on the Tor network. Privacy and anonymity are crucial for ensuring freedom of expression on the internet. Privacy ensures that no entity can gather personal or private information about a user, while anonymity protects all information that could reveal the user's true identity (Mehta and Upadhyay, 2020; Tavisionabi et al., 2019). It is worth noting that the complexities surrounding secure and efficient communication are not confined solely to the domain of web networks. In parallel fields such as Vehicle-to-Vehicle (V2V) communication, a variety of technologies—including but not limited to Lifi, RFID, VANET, and LORAWAN—are under rigorous examination for their potential to enhance road safety and optimize traffic management (Agarwal et al., 2021; Agarwal and

Sharma, 2022a). Analogous to the challenges faced in the Tor network, these technologies must also navigate intricate issues such as ensuring secure, real-time data transmission amidst concerns of data imbalance and obfuscation (Agarwal and Sharma, 2022b; Agarwal et al., 2023).

The Tor network, initially developed by the U.S. Navy in the 1990s for secure government communications, has evolved into an open-source platform facilitating anonymous communication (Omar and Ibrahim, 2020). Its early adoption was largely driven by privacy advocates, activists, and journalists. Over time, the network has expanded its appeal, particularly as public awareness of online privacy and surveillance has increased. This broader usage has led to the introduction of “hidden services” in the early 2000s, allowing users to host anonymous websites and services within the network. These features have been embraced by a diverse user base, including those involved in underground economies (Biryukov et al., 2013). With growing concerns over privacy, the Tor browser has emerged as the preferred choice for users seeking robust encryption and complete anonymity. However, this high level of anonymity has a double-edged nature, offering cover for illicit activities and thus posing challenges for law enforcement agencies. To further enhance user privacy, the network has recently implemented next-generation onion services, improving both the performance and

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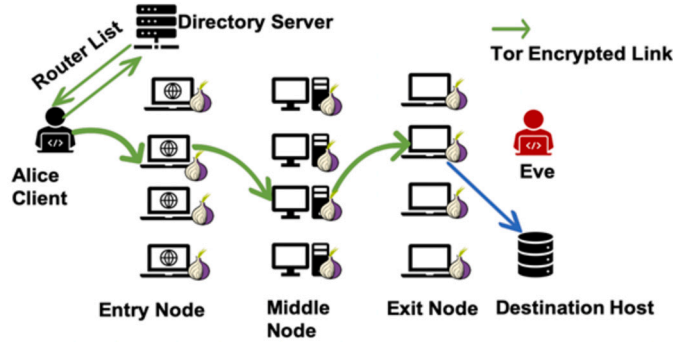


Fig. 1. Tor Network Traffic (Johnson et al., 2021).

security of hidden services. In addition, Tor has experienced a surge in usage among individuals in countries with restrictive internet policies, providing them a means to access blocked content and services (Harborth et al., 2019).

Furthermore, underpinning these features is Tor's technology, first introduced by the U.S. Naval Research Laboratory in 2004. The architecture of this system, illustrated in Fig. 1, is designed for maximum data concealment. Data enters the network at an entry node and is encrypted and relayed through multiple internal nodes before exiting through an exit node. This labyrinthine routing structure further complicates efforts to detect Tor traffic, presenting significant obstacles for the identification of criminal activities (Bian et al., 2021; Johnson et al., 2021).

The dual nature of Tor and its associated tools like Obfs4 and meek make it a complex but essential technology (Xu and Zou, 2021). While it is true that such platforms can be exploited for illicit purposes, their role in preserving the anonymity and freedom of human rights defenders, journalists, and vulnerable populations is undeniable (Thorsen, 2019; Lewman, 2013; Bell and Owen, 2017). The usage of these tools highlights their global importance. With so many relying on Tor and its associated technologies to safely access information, any research that exposes vulnerabilities should be shared responsibly with the Tor Project first. By doing so, the organization can assess the potential risks and work on mitigation strategies that could protect hundreds of thousands of users worldwide who have no other secure way to access or share information.

The use of Tor in establishing secure channels for whistleblowers. Given the increasingly precarious nature of journalism and activism in various parts of the world, these channels are sometimes the only way to expose malpractice, corruption, or human rights abuses (Di Salvo, 2021). The use of Tor for whistleblower tip lines by major news organizations further underlines its value in today's society. It's about ensuring that such scrutiny leads to stronger, more secure systems rather than putting already vulnerable communities at greater risk. Responsible disclosure offers a path that respects the technology's multifaceted role in modern society.

The inherent complexity of this routing architecture increases the existing challenges associated with detecting illicit activities on the Tor network. These challenges include the constant evolution of obfuscation techniques, a reliance on a limited feature set, the risk of model overfitting, high computational overhead, a lack of publicly available datasets, and an incomplete understanding of Tor traffic patterns and characteristics. These gaps make it difficult to accurately identify and block Obfuscated Tor traffic and require careful trade-offs between accuracy, performance, and computational cost in detection methods. In addition, existing literature highlights the limitations in the efficacy of current detection methods for identifying obfuscated Tor traffic, especially when faced with advanced obfuscation techniques. Furthermore, the lack of robustness in established detection models makes them vulnerable to adversarial attacks and evasion strategies. There is also a notable deficiency in the emphasis on real-time detection and scalabil-

ity, both of which are critical for handling large-scale network traffic. On the other hand, Deep Learning (DL) models are crucial in detecting obfuscated Tor traffic as they can help address the limitations of traditional methods. These techniques can help in detecting evolving obfuscation techniques, detecting limited feature sets, reducing overfitting, being more computationally efficient, dealing with limited public datasets, and understanding the patterns and characteristics of Tor traffic. Overall, DL models can improve the accuracy and efficiency of detecting obfuscated Tor traffic by overcoming the limitations of traditional methods. The contributions of this paper include:

- A proposed detection approach that integrates a Bidirectional Generative Adversarial Network (BiGAN) and a Vision Transformer (ViT) model to enhance obfuscated Tor-nonTor traffic detection.
- Data augmentation approach to address the high-class imbalance problem in detecting Tor-nonTor traffic through generating a synthetic Tor-nonTor (IS-CXTor2016) dataset using a BiGAN model.
- Train a ViT model using synthetic and original datasets to multi-obfuscated Tor-nonTor detection.
- Performance evaluation of the proposed approach and analysis of the effectiveness of the approach on balanced and imbalanced data. Moreover, the findings indicate that the proposed outperforms other state-of-the-art approaches.

This paper is structured as follows: Section 2 provides an overview of the prior works about identification traffic Tor. Section 3 illustrates the methodology of this paper, and the results discussed in Section 4. Finally, the conclusion in Section 5.

2. Related works

There has been a growing body of research on the detection of obfuscated Tor traffic using deep learning techniques. One approach is the use of convolutional neural networks (CNNs) for feature extraction. For example, in the work of Rust-Nguyen and Stamp (2022), the authors proposed a CNN-based method for detecting obfuscated Tor traffic. They used CNN to extract features from the network traffic and employed a Support Vector Machine (SVM) to classify the traffic as Tor or non-Tor. The proposed method achieved high accuracy in detecting obfuscated Tor traffic. Another approach is the use of Recurrent Neural Networks (RNNs) for feature extraction. Moreover, the study in Sarkar et al. (2020) introduced a flexible approach called AI-FlowDet, a CNN-based flow-splitting method that leverages the concept of scene change detection. The method demonstrated notable improvements in flow application type classification. Besides that, AI-FlowDet utilized the UNB dataset for its experiments and introduced a comprehensive set of 294 size-based and direction-based features specifically designed for application-type characteristics (Chen and Lin, 2021).

In the work of Wang and Chen (2019), the authors proposed an RNN-based method for detecting obfuscated Tor traffic. They used a long short-term memory (LSTM) network to extract features from the network traffic and then used a Decision Tree (DT) to classify the traffic as Tor or non-Tor. The proposed method achieved high accuracy in detecting obfuscated Tor traffic. A third approach is the use of transfer learning to improve the performance of detection. In this study (Sarkar et al., 2020), the authors proposed a transfer learning-based method for detecting obfuscated Tor traffic. They used a pre-trained Deep Neural Network (DNN) model as a feature extractor and then fine-tuned the model on a small dataset of Tor traffic. The proposed method achieved high accuracy in detecting obfuscated Tor traffic.

Furthermore, Twelve features were extracted from packets using a sliding window technique by Xu and Zou (2021). The obfuscated Tor traffic was classified into three protocol obfuscation types: Meek, Obfs4, and FTE, using seven machine learning models: Extreme Gradient Boosting (XGBoost), Gradient Boosting Decision Tree (GBDT), Random Forest (RF), Classification and Regression Trees (CART) decision

tree, K-Nearest Neighbor (KNN), Logistic Regression (LR), and SVM. The RF model achieved a high precision rate, but there was a slight gap in detecting Meek-based and Obfs4-based Tor traffic compared to detecting only one type of obfuscated Tor traffic. While the study in He et al. (2019) presented two-tiered approach for the high-precision, real-time recognition of Obfs4 traffic. Initially, a coarse-grained filtering phase employs a randomness detection algorithm focused on the handshake packet payload. This phase also leverages the timing sequence characteristics of packets in the handshake process to eliminate other types of traffic that could interfere with the analysis. Following this, the research moves into a fine-grained identification phase where it utilizes statistical features derived from a large dataset of Obfs4 traffic. Various classification algorithms were tested, with the SVM model demonstrating an accuracy rate above 99% for identifying Obfs4 traffic. These findings suggest that Obfs4 may not be entirely effective in countering traffic analysis attacks in real-world applications.

Okonkwo et al. (2022) proposed a CNN model with eleven layers to classify the encrypted network traffic in HTTPs, VPN, and Tor. The model transformed network packets into unique images for each traffic class, based on packet size and arrival time features. The proposed model addressed over- and under-fitting issues but used a simple technique to deal with the imbalance issue. However, the model achieved higher accuracy for VPN traffic than Tor traffic, and it was not able to classify audio and chat traffic accurately. Zhao et al. (2019) presented the MT-DNN-FL model, which stands for Multi-Task Deep Neural Network in Federated Learning (FL), to detect anomaly traffic in VPN and Tor. The paper compared this model with single-task models such as DNN, C4.5 DT, LR, KNN, and RF. The MT-DNN-FL model consisted of three layers: (i) Input layer, (ii) Shared layers, which performed feature extraction and multi-task settings in DNN, and (iii) Task layers, which comprised subnetworks for each specific task. Although the study discovered that the multi-task model surpassed the single-task model in DNN performance, the employment of an imbalanced dataset in the experiment may have influenced the model's effectiveness. Besides that, optimizing the DNN while maintaining training performance presents potential challenges, which could be arduous or require an extended duration to achieve.

Furthermore, the study conducted in Abu Al-Haija et al. (2022) explored and assessed six supervised machine learning methods employed in Darknet Traffic Detection Systems (DTDS) specifically designed for IoT networks. The techniques encompass bagging decision tree ensembles (BAG-DT), AdaBoost Decision Tree ensembles (ADA-DT), RUS-Boosted Decision Tree ensembles (RUS-DT), Optimizable Decision Tree (O-DT), Optimizable K-Nearest Neighbor (O-KNN), and Optimizable Discriminant (O-DSC). Among these methods, BAG-DT demonstrated superior performance relative to the other techniques by effectively classifying four categories that include VPN, non-VPN, Tor, and non-Tor traffic. While in Marim et al. (2023), the authors addressed the challenges of detecting and characterizing Darknet traffic using decision tree-based learning models, including DT, RF, and Multilayer Perceptron (MLP), which achieve high classification accuracy. Moreover, the paper utilized the Recursive Feature Elimination (RFE) algorithm to select the most relevant features for model prediction. Although simple machine learning algorithms, such as decision tree-based models, can provide competitive results for real-world problems, they may face issues related to real-time prediction.

Table 1 provides a summary of related works, outlining the reference, model, benefits, and limitations of each study in the context of our research.

The main research gaps in the existing works on the detection of obfuscated Tor traffic using deep learning techniques include lack of evaluation on a large dataset, lack of robustness against new and evolving obfuscation techniques, lack of comparison with other state-of-the-art methods, and lack of evaluation of computational cost. In general, these works do not sufficiently evaluate their proposed methods against

real-world scenarios and other current techniques, leaving room for further research in this area.

3. Methodology

The main emphasis of this section is on elaborating on the proposed approach for detecting obfuscated traffic. The approach comprises four primary phases: (i) analyzing and comprehending the dataset, (ii) pre-processing the dataset, (iii) addressing the imbalance issue using data augmentation via the BiGAN model, and (iv) detecting obfuscated traffic using the ViT model, as illustrated in Fig. 2.

3.1. Dataset description

ISCXTor2016 dataset is a collection of network traffic data that was gathered by the Information Security Center of Excellence (ISCX) at the University of New Brunswick in Canada in 2016. The dataset contains both normal and Tor traffic and consists of approximately 2.5 million flows captured over a period of 2 weeks. These flows were captured from various network interfaces and devices such as laptops, smart-phones, wired and wireless interfaces, and servers (Palmieri, 2019; Vasan and Surendiran, 2016; Jiang et al., 2022). The dataset is labeled, it includes normal traffic, primarily from web browsing, file transfer, and email activities, and Tor traffic which can include both legitimate and malicious activities like file sharing, anonymous communication, and command and control of malware. ISCXTor2016 dataset is commonly used in research on the detection of Tor traffic as it provides a large and diverse set of data that can be used to train and evaluate detection methods (Zhao et al., 2019; Lin et al., 2021). However, it's important to keep in mind that the dataset is not recent, and the Tor network and its usage may have changed since 2016.

The ISCXTor2016 dataset, also known as the Tor-nonTor dataset, is divided into two scenarios, A and B, based on time features and includes five different flow-timeout values: 10 s, 15 s, 30 s, 60 s, and 120 s. Scenario A is a binary classification task that distinguishes traffic between Tor and NonTor, as presented in Table 2. However, Scenario A has achieved high-performance results in prior works, such as in Johnson et al. (2021); Okonkwo et al. (2022).

On the other hand, Scenario B is to find out the type of Tor traffic as one of eight categories (multi-classification). Moreover, Scenario B is still suffering from the slight performance hit of multi-classification. Table 3 presents eight classes for Tor traffic (Lashkari et al., 2017; Yin et al., 2022).

Indeed, Tor traffic is related to TCP protocol where it does not support UDP protocol, thus ISCXTor2016 dataset has TCP traffic with bidirectional flows: forward packet (source to destination) and backward packet (destination to source). Ultimately, there are 23 features that calculated based on statistical time-related features have been provided in the ISCXTor2016 dataset, and the detail in Table 4 (Lashkari et al., 2017). In particular, Active Mean, Active Std, Active Max, Active Min, Idle Mean, Idle Std, Idle Max, and Idle Min features have been eliminated, because the most value of these features is zero which leads to noise for Obfuscated traffic detection. The next section presents the data preprocessing in detail.

3.2. Data preprocessing phase

Preprocessing is a significant stage in preparing the data and acquiring accurate results in the next stage of the proposed model. However, there are several steps for preprocessing, but it depends on the raw data (Al-E'mari et al., 2021). In the proposed model the following preprocessing steps on the ISCXTor2016 dataset:

- Data Cleaning: The noise data leads to incorrect results; therefore, to improve the accuracy of the proposed model, all Infinity, NaN, and missing value rows have been eliminated from the dataset.

Table 1
Summary of related works.

Ref	Model	Benefits	Limitations
Rust-Nguyen and Stamp (2022)	CNN+SVM, RNN	The method achieves high accuracy in detecting obfuscated Tor traffic. Besides, the model can automatically extract relevant features from the network traffic by the CNNs model.	The performance of the model might be contingent upon the selection of hyperparameters, a process that could entail substantial computational resources and temporal investment for optimization.
Chen and Lin (2021)	CNN	The method was tested and proved effective for both TOR and VPN traffic, indicating its broader applicability.	Although the feature set is comprehensive, it is static and may not adapt well to new types of traffic or obfuscation techniques that could emerge in the future.
Wang and Chen (2019)	RNN, LSTM+DT, Transfer learning	LSTM has overcome the challenges in RNN, and DT is easy to implement. Furthermore, the transfer learning approach facilitates the employment of pre-trained models, thereby mitigating the necessity for substantial computational resources and minimizing the duration dedicated to training.	LSTM has challenges of vanishing and exploding gradients throughout the training process, potentially rendering them less apt for extensive or real-time applications. While the effectiveness of transfer learning largely relies on the similarity between source and target tasks; significantly different tasks may result in minimal or even harmful performance improvements.
Sarkar et al. (2020)	Transfer learning	Employing transfer learning techniques may enhance the efficacy of detection by capitalizing on insights gleaned from pertinent, associated tasks. This approach mitigates the necessity for copious quantities of labeled data and holds the potential to expedite the training process.	The efficacy of transfer learning is contingent upon the degree of congruence between the source and target tasks; in certain instances, substantial performance enhancements may not be realized. Furthermore, this method may necessitate supplementary fine-tuning, which could prove to be laborious and time-intensive.
Xu and Zou (2021)	XGBoost, GBDT, RF, CART, KNN, LR, SVM	The paper encompasses three distinct protocol obfuscation types, namely Meek, Obfs4, and FTE, thereby fostering a more comprehensive comprehension of obfuscated Tor traffic detection. Furthermore, the employment of seven disparate machine learning models facilitates a comprehensive assessment and comparison of their efficacy in the identification of obfuscated Tor traffic.	There was a slight gap in detecting Meek-based and Obfs4-based Tor traffic compared to detecting only one type of obfuscated Tor traffic. In addition, the limited evaluation metrics would provide a more comprehensive understanding of the models' performance.
He et al. (2019)	SVM	The study used a coarse-grained filtering phase to help in the quick elimination of irrelevant traffic and enhance the efficiency of the method.	The study concentrates solely on Obfs4 traffic, limiting its applicability to other types of obfuscated or encrypted traffic.
Okonkwo et al. (2022)	CNNs	The model with eleven layers is capable of extracting high-level features from the data, potentially improving classification performance. Furthermore, it mitigated overfitting and underfitting issues.	The model utilizes a relatively simple technique to address the imbalance issue, which may not be as effective as more sophisticated approaches.
Zhao et al. (2019)	MT-DNN-FL	The MT-DNN-FL method ensures data privacy by not sharing training data with third parties and performs multiple tasks concurrently, offering network administrators extensive insights into network anomalies, VPN (Tor) traffic, and traffic classification.	The MT-DNN-FL framework may exhibit complexity, eliciting concerns pertaining to its scalability, interpretability, and practicability within authentic contexts, where the current evaluation of the proposed method is restricted to specific domains.
Abu Al-Haija et al. (2022)	BAG-DT, ADA-DT, O-DT, O-KNN, O-DSC	The paper evaluates the effectiveness of six supervised machine-learning methods, providing a diverse set of methods for evaluation. Moreover, the proposed model can be efficiently deployed to detect Tor and VPN activities in communication networks, demonstrating its real-world applicability.	The paper focuses exclusively on supervised machine learning techniques, potentially overlooking unsupervised or semi-supervised learning methods. Moreover, although the CIC-Darknet-2020 dataset used for assessment is modern and comprehensive, it may not capture all possible variations in IoT network traffic or future emerging threats.
Marim et al. (2023)	DT, RF, MPL	The paper utilized simple models for traffic prediction, extracted and generated new features, and adapted RFE for feature selection. In addition, it highlighted the effectiveness of the models in detecting and characterizing Darknet traffic.	Although the MLP model is suitable for real-time traffic analysis, its lower performance compared to DT and RF may hinder its adoption in certain applications, as well as its poor performance with unbalanced labels in the dataset.

Table 2
Tor-NonTor Network Traffic Dataset Content for Scenario A.

Time	Tor	NonTor	Total
10 s	8044	59790	67834
15 s	5631	48123	53754
30 s	3130	43892	47022
60 s	1723	41376	43099
120 s	1069	38285	39254

Furthermore, tear out the duplicate rows from the dataset (Arikkat et al., 2022).

- **Data Transformation:** It converts the format of the data into pure and usable data. The proposed model applied the data transformation in two ways, it replaces words with numerical values as shown Id column in Table 4, and it converts the network traffic into a NumPy array.

- **Data Normalization:** It is utilized in order to handle outliers in the dataset and enhance the results. The proposed model re-scales data between 0 and 1 through the min-max technique by Equation (1) (Al-E'mari et al., 2021).

$$f(x) = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (1)$$

- **Data Merging:** This step involves combining Tor and non-Tor traffic into a single file in preparation for training the proposed model. There are five files related to flow timeout (10 s, 15 s, 30 s, 60 s, and 120 s), each of which has nine classes as presented in Table 5.
- **Data Shuffling:** The dataset is properly shuffled before training the model, it is a significant step since avoids any element of bias/pattern in splitting the dataset to training, validation, and testing set (Kakinaka and Umeno, 2022).
- **Data Visualization:** This step converts the traffic to NumPy format, and then data visualization can be presented with 224x224 pixel images and saved as PNG files. Fig. 3 illustrates a sample of the Non-Tor and Tor traffic as images, the index for each image related to Table 5.

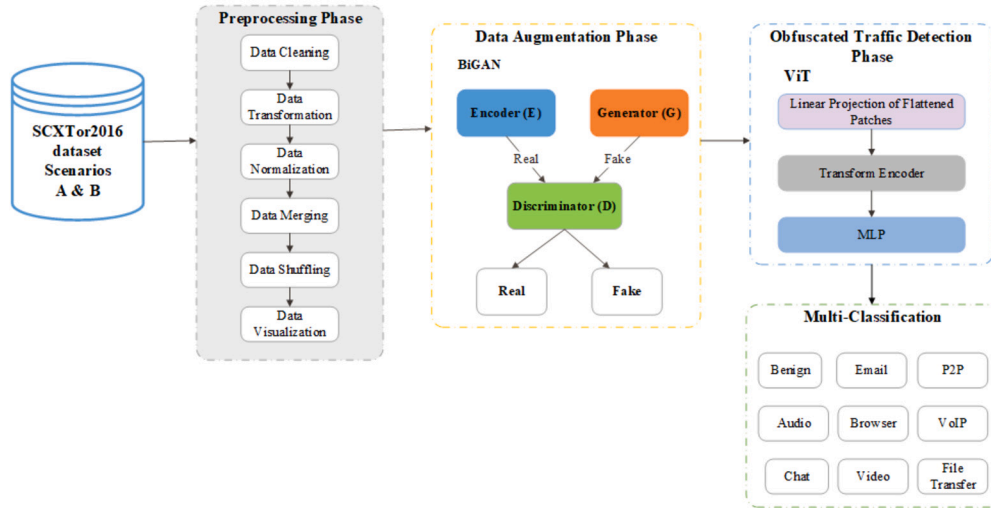


Fig. 2. The Workflow of the proposed approach.

Table 3
Tor Network Traffic Dataset Content for Scenario B.

Time	Browser	Email	Chat	Audio	Video	File Transfer	VoIP	P2P	Total
10 s	1604	282	323	721	874	864	2291	1085	8044
15 s	1194	194	249	510	617	590	1544	733	5631
30 s	694	111	153	332	364	311	790	375	3130
60 s	411	60	90	190	196	165	413	198	1723
120 s	239	34	151	119	105	86	225	110	1069

Table 4
Features description.

Feature	Description
Flow duration	Length of the connection (seconds)
Flow bytes	Number of bytes sent
Flow packets	Number of packets sent
Flow IAT	Packets flow inter-arrival time (max, min, mean, std)
Fwd IAT	Forward inter-arrival time (max, min, mean, std)
Bwd IAT	Backward inter-arrival time (max, min, mean, std)
Active	Seconds in which the flow has been active (max, min, mean, std)
Idle	Seconds in which the flow has been idle (max, min, mean, std)

Table 5
Tor-NonTor Network Traffic Dataset Content after Preprocessing Stage.

Id	Class	10 s	15 s	30 s	60 s	120 s
0	Benign	59784	37516	29302	28701	21564
1	Audio	721	92	64	45	32
2	Browser	1604	454	266	152	82
3	Chat	323	486	294	166	90
4	File Transfer	864	960	492	246	126
5	Email	282	372	208	105	56
6	P2P	1085	142	76	38	20
7	Video	874	1196	690	343	180
8	VoIP	2291	3018	1516	756	386

3.3. Data augmentation phase

The dataset has been analyzed after the preprocessing phase, where the data suffers from an imbalance issue. It turns out that normal data is extensive in contrast to other data is slight as shown in Fig. 4. To overcome this issue the proposed model utilizes the BiGAN model to generate data and produce balance data then increased detection accuracy. The survey conducted in Shorten and Khoshgoftaar (2019) on data augmentation techniques for deep learning revealed that accuracy can

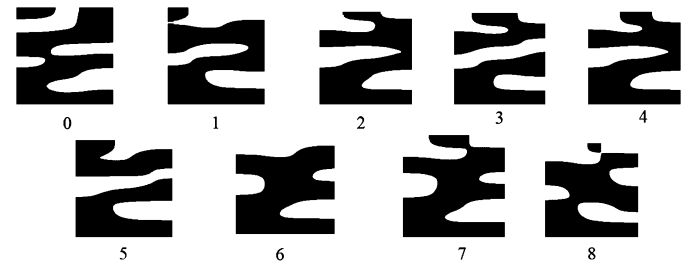


Fig. 3. Snapshot of Data Visualization Step.

be improved by utilizing geometric transformations such as flipping, cropping, rotating, and color space transformations.

To address data imbalances in training and testing, heavy data augmentation was applied to the images. Therefore, in the proposed model, data augmentation was implemented using a BiGAN on image data generated from the previous preprocessing phase. The BiGAN architecture comprises three primary components: (i) Encoder, which is a neural network that maps an input data sample to a low-dimensional representation or embedding; (ii) Generator, which is a neural network that generates synthetic data samples similar to the original data from the latent code (Kaplan and Alptekin, 2020; Xu et al., 2022). The generator is trained to produce synthetic samples that cannot be distinguished from real data samples by the discriminator; and (iii) Discriminator, which is a neural network that evaluates the authenticity of the generated samples. The discriminator is trained to differentiate between real and synthetic data samples and to reconstruct the original data sample from its embedding. Fig. 5 illustrates the architecture of BiGAN (Yang and Li Bigan, 2021; Ding et al., 2020).

In BiGAN, an encoder (E) component has been added to the typical GAN structure, transforming the input data into a latent space representation. The discriminator (D) receives both the latent representation from the encoder and the input data as its input. This differs from stan-

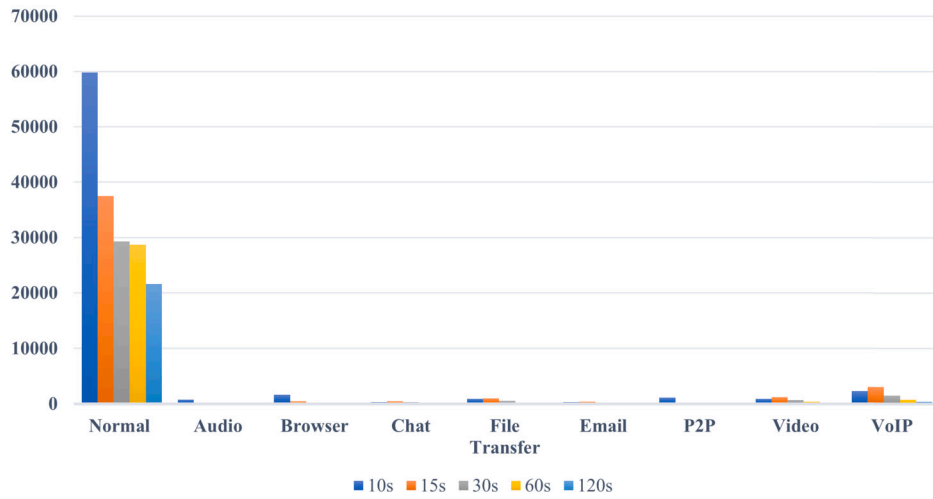


Fig. 4. Dataset before Data Augmentation.

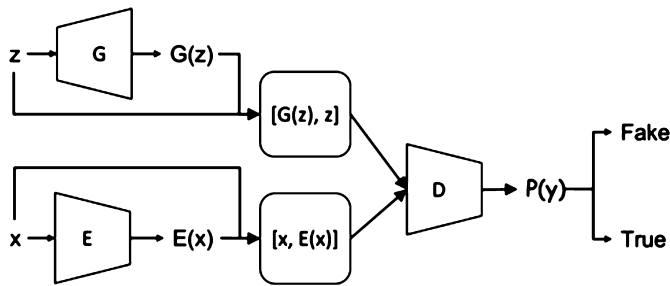


Fig. 5. Architecture of BiGAN (Xu et al., 2022).

Table 6
Synthesized data of the ISCXTor2016 Dataset.

Id	Class	10 s	15 s	30 s	60 s	120 s
0	Benign	49784	37516	29302	28701	215644
1	Audio	47210	32000	29640	28450	21320
2	Browser	36040	35400	29266	28152	21820
3	Chat	32300	34860	29294	28166	21900
4	File Transfer	36400	29600	29492	28246	21126
5	Email	38200	33720	29208	28105	21560
6	P2P	40850	31420	28760	28380	21200
7	Video	37400	31196	26900	28343	21180
8	VoIP	42910	33018	21516	28756	21386

dard GANs, where the discriminator only considers the generated data (G). The loss function for BiGAN is defined in Equation (2).

$$\min_{G,E}, \max_D = -E_{X \sim p_X} [E_{Z \sim p_Z} (\log D(X, Z))] + [E_{Z \sim p_Z} [E_{X \sim p_{G(Z)}} [\log(1 - D(x, z))]] \quad (2)$$

In this equation, X represents real data, G(z) denotes fake data, and P symbolizes the density of data. To elaborate, the right operand represents the expectation of the log probability that the D correctly identifies the real data samples (x) and their corresponding latent codes (Z) generated by the (E) from the true data distribution(p_x). On the other hand, the left operand represents the expectation of the log probability that the (D) correctly identifies the generated data samples (X) and their corresponding latent codes (Z) from the (G) as fake (i.e., not real) (Donahue et al., 2016). However, the BiGAN model has generated data for five files to overcome the imbalance issue and enhance the results of the detection model. The number of synthesized data of the ISCXTor2016 dataset is demonstrated in Table 6.

In addition, Fig. 6 is a visualization that demonstrates the impact of the data augmentation process using the BiGAN model on the original dataset. Besides that, it should be noted that the data in Fig. 6 is balanced compared to Fig. 4.

3.4. Obfuscated traffic detection phase

The ViT model is a type of deep learning model that was introduced for image classification tasks in 2020 by Dosovitskiy et al. ViT was originally developed for Natural Language Processing (NLP) tasks where the main idea behind ViT is to convert an image into a sequence of patches and then process them using a transformer-based architecture as shown in Fig. 7. The transformer model is able to capture long-range dependencies between the patches and produce a fixed-length feature vector that represents the image. Moreover, the ViT model has demonstrated impressive performance on several image classification benchmarks and has been noted for its ability to handle large-scale image recognition tasks while being computationally efficient (Dosovitskiy et al., 2020; Li et al., 2023).

The architecture of ViT is designed to process the input image patches and learn complex relationships between them using self-attention mechanisms in the transformer layers. However, ViT does involve a form of feature extraction in the sense that it extracts image patches from the input image and converts them into embedding vectors, which are then processed by the transformer layers. These embedding vectors can be considered low-level features, and the transformer layers learn higher-level representations of these features (Paul and Chen, 2022; Chen et al., 2022).

4. Experiments

In this section, the results of the proposed model for the detection of obfuscated Tor traffic using a combination of BiGAN and ViT are presented. Furthermore, this section conducts an in-depth analysis of the results to understand the factors that contribute to the performance of the proposed approach and investigate the impact of different types of obfuscation techniques on the accuracy of the model.

4.1. Evaluation metrics

In this section, the proposed model will be evaluated for its ability to detect obfuscated Tor traffic using standard metrics for network anomaly detection and network traffic classification. These metrics include **Accuracy**, which serves as a means to assess the performance of classification models by quantifying the ratio of correct predictions, encompassing both true positives and true negatives, to the overall

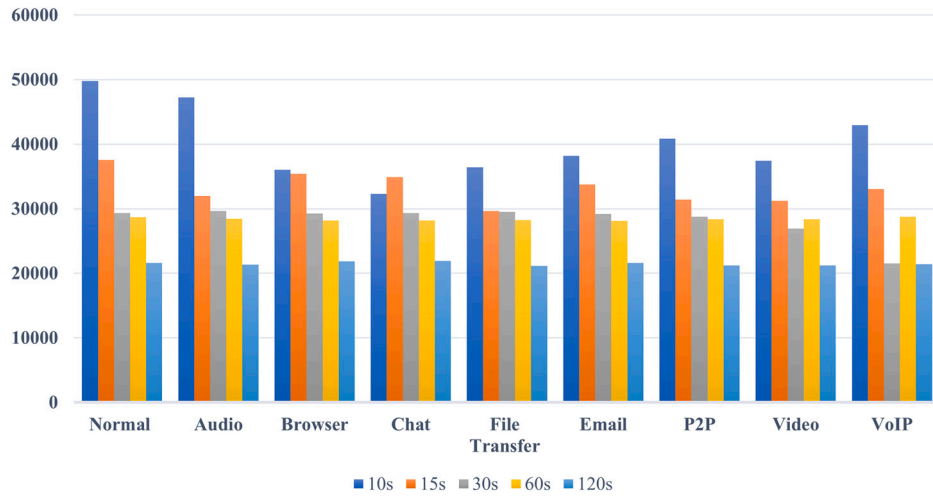


Fig. 6. Dataset after Data Augmentation.

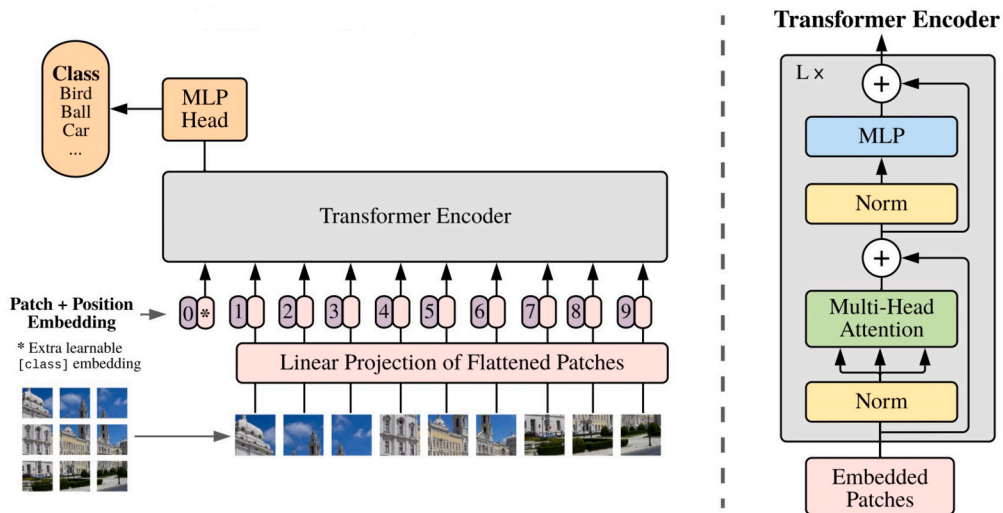


Fig. 7. Architecture of ViT (Yang and Li Bigan, 2021).

number of predictions made, as shown in Equation (3), **Recall**, which represents the proportion of correctly identified Tor traffic out of all actual Tor traffic instances within Equation (4), and **Precision**, which measures the proportion of true Tor traffic instances among those labeled as Tor by the model within Equation (5) (Al-E'mari et al., 2021).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$Presion = \frac{TP}{TP + FP} \quad (5)$$

Furthermore, **F1-Score** metric to evaluate the performance of the proposed model, combines two metrics (recall and precision) into a single value, which ranges from 0 to 1, with 1 being the best possible score. It is defined as (Paul and Chen, 2022):

$$F1 - Score = 2 * \frac{Presion * Recall}{Presion + Recall} \quad (6)$$

The evaluation metrics are derived from the confusion matrix, as illustrated in Table 7. The term True Positive (TP) denotes the instances where the model correctly identifies the traffic. Conversely, False Positive (FP) represents the occurrences when the model predicts a positive outcome while the actual class is negative. True Negative (TN) signifies the instances where the model accurately predicts a negative outcome,

Table 7
Confusion matrix.

	Predict		
		Positive	Negative
Actual	True	True Positive (TP)	True Negative (TN)
	False	False Positive (FP)	False Negative (FN)

and False Negative (FN) indicates the situations where the model incorrectly predicts a negative outcome when the actual class is positive (Al-E'mari et al., 2022; Sanjalawe and Althobaiti, 2023; Althobaiti et al., 2023).

4.2. Results and discussion

This section demonstrates the effectiveness and potential of the proposed approach for detecting obfuscated Tor traffic and provides insights into the factors that influence its performance by comparing ViT with Recurrent Neural Networks RNNs and CNNs training and test models on an original and synthetic dataset. Furthermore, the experimental results in Table 8 show that the proposed approach is able to accurately detect obfuscated Tor traffic with high precision and recall, outperforming existing state-of-the-art methods.

Table 8
Result summary comparison.

Ref	Model	Accuracy	Recall	Precision	F1-score
Wang and Chen (2019)	LSTM	83.42%	88.89%	82.76%	85.72%
Xu and Zou (2021); Marim et al. (2023)	RF	87.57%	92.37%	87.90%	90.08%
Xu and Zou (2021)	XGBoost	84.98%	92.13%	86.03%	88.97%
Xu and Zou (2021); Zhao et al. (2019)	KNN	82.91%	90.00%	85.40%	87.64%
Rust-Nguyen and Stamp (2022); Wang and Chen (2019)	RNN	86.53%	92.24%	86.29%	89.17%
Rust-Nguyen and Stamp (2022); Okonkwo et al. (2022)	CNN	89.64%	95.00%	89.06%	91.94%
Wang and Chen (2019); Marim et al. (2023)	DT	90.16%	95.04%	89.84%	92.37%
Marim et al. (2023)	MLP	90.16%	95.04%	89.84%	92.37%
The Proposed Approach	ViT	92.23%	96.53%	93.29%	94.88%
	RNN+BiGAN	94.30%	96.77%	96.15%	96.46%
	CNN+BiGAN	98.45%	99.72%	98.62%	99.17%
	ViT+BiGAN	99.59%	99.83%	99.72%	99.78%

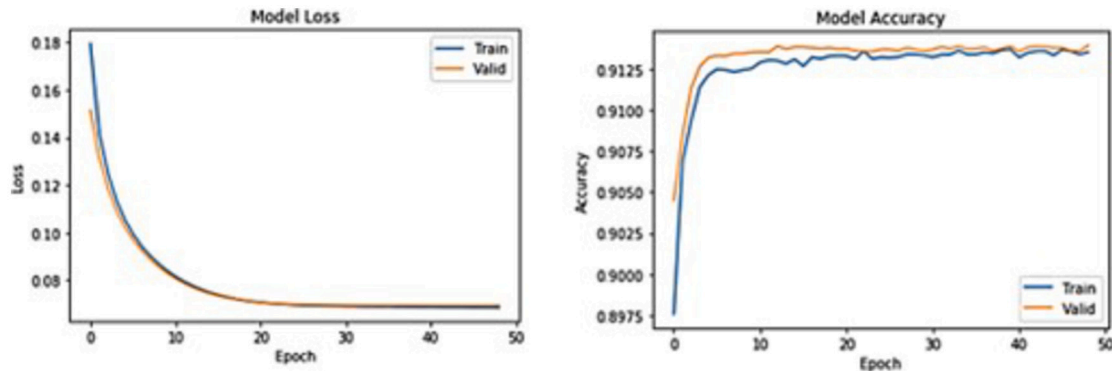


Fig. 8. ViT+BiGAN learning curves for model loss and model accuracy.

Table 8 illustrates the average results of the SCX-Tor2016 dataset in terms of accuracy, recall, precision, and F1-score for five different file data based on flow-timeout values (10 s, 15 s, 30 s, 60 s, and 120 s). In addition, the ISCX-Tor2016 dataset was divided into subsets for training, validation, and testing with 70%, 10%, and 20% respectively. As well as the training models executed after the preprocessing phase then compare the results before and after data augmentation.

In the BiGAN-ViT approach, hyperparameter tuning involves several key parameters. Namely: (i) the loss function is set to categorical cross-entropy, (ii) the activation function is chosen to be the softmax function, (iii) the optimizer used for updating the parameters is Adam, (iv) the learning rate is set to 10^{-4} , (v) the batch size is 32, and the model is trained for various numbers of epochs to determine the optimal training duration. Moreover, the hardware specifications employed for training the BiGAN-ViT approach included a computationally robust setup, utilizing the capabilities of a high-performance NVIDIA GeForce RTX 3090 GPU, which significantly accelerated the training process. The system was equipped with 16 GB of RAM to accommodate large datasets and intermediate computations. A 1TB Solid State Drive (SSD) was also employed to facilitate rapid data access and efficient storage of the dataset, model weights, and other relevant files.

As a result, Fig. 8 demonstrates the learning curve for the proposed model (ViT+BiGAN) where the model loss decreases over time for both the training and validation sets, which indicates that the model is learning well. The model accuracy is increasing over time for both the training and validation sets that refer to learning perfectly.

From the empirical results, the ViT+BiGAN model has the highest accuracy of 99.59%, which means that it correctly identified almost all of the samples in the dataset. On the other hand, all models have relatively high recall, precision, and F1-score values, with ViT+BiGAN having the highest recall of 99.83%, precision of 98.62%, and F1-score of 99.78%. Overall, the experimental results show that the models using the BiGAN approach (RNN+BiGAN, CNN+BiGAN, ViT+BiGAN) outperformed the models without the BiGAN approach (RNN, CNN, ViT) in

terms of accuracy, recall, precision, and F1-score. Besides, the ViT+BiGAN model performed the best overall, with high scores in all metrics. However, KNN, XGBoost, and LSTM models have the lowest accuracy of 82.91%, 84.98%, and 83.42%, respectively. As well as the other metrics have lower performance compared with RNN+BiGAN, CNN+BiGAN, and ViT+BiGAN models.

The combination of BiGAN and ViT, two deep learning architectures, has exhibited the potential to improve the performance of Intrusion Detection Systems (IDS). BiGAN is a generative adversarial network that can learn both the generator and discriminator networks bidirectionally, resulting in the capture of complex and high-dimensional data distributions. This property has made BiGAN a suitable candidate for learning the underlying data augmentation to increase the network traffic and assist other models in detecting traffic patterns, thereby increasing the detection rates, reducing the false positive rates of IDS, and overcoming the overfitting issue. Similarly, ViT is a class of deep neural networks that can capture long-range dependencies in image data by utilizing self-attention mechanisms. ViT has demonstrated remarkable performance in image classification tasks and has been extended to IDS, where ViT-based models can learn to extract features from network traffic and detect anomalies in traffic patterns. The self-attention mechanism in ViT-based IDS models facilitates the capture of the dependencies between network packets and the identification of patterns associated with malicious activities. Furthermore, researchers have explored the combination of BiGAN and ViT in IDS, which has shown promise in achieving even higher accuracy and detection rates by using BiGAN to learn the underlying data distribution and ViT to extract relevant features from the network traffic. Fig. 9 depicts a visual representation of the experimental results obtained in this paper.

In sum, vision transform technologies have the potential to greatly enhance IDS by improving their accuracy, speed, and flexibility. By utilizing computer vision algorithms to analyze video footage from surveillance cameras, IDS can accurately detect and track the movement of intruders in real-time, leading to quicker response times and

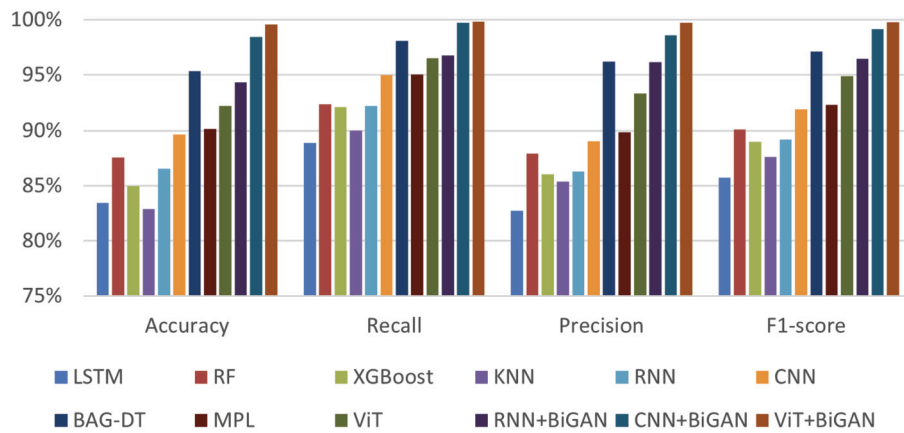


Fig. 9. Performance evaluation.

reduced false alarms. In contrast to traditional IDS, which may trigger false alarms due to changes in light or environmental factors, vision-based IDS can filter out these false positives and only alert security personnel to actual intrusions. Furthermore, vision-based IDS can be integrated with other security systems such as access control and fire alarms, providing a comprehensive security solution for organizations. Overall, the use of vision transform technologies in IDS can significantly improve security and safety measures for people and property.

On the other hand, BiGAN can play a crucial role in improving the performance of intrusion detection systems (IDS) by generating synthetic data that can be used for training and testing purposes. BiGAN is a type of GAN that can generate data in both forward and backward directions, mimicking the input features and output labels of real-world data. This synthetic data can supplement the limited real-world data available for training IDS, resulting in better accuracy, faster detection, and fewer false alarms. By generating synthetic data that resembles real-world attacks, BiGAN can train IDS to detect and respond to different types of attacks more effectively. Furthermore, BiGAN can augment real-world data by generating new instances of it, which can improve the diversity and complexity of the training data, leading to better detection of unknown attacks. Overall, the use of BiGAN in IDS can improve the quality and quantity of training data, leading to better accuracy and reliability in intrusion detection. This can result in improved security outcomes, which is critical in ensuring the safety and protection of people and property.

5. Conclusions

Tor is widely used for hiding criminal activities on the darknet, making it crucial to be able to detect and block such traffic. However, Tor has implemented various obfuscation techniques to evade traditional detection methods, leading to challenges in identifying such traffic. In this research, a BiGAN and ViT approach for detecting Obfuscated Tor Traffic is proposed. The characteristics of different obfuscation techniques were analyzed, and their effectiveness was evaluated using the ISCX-Tor2016 dataset. The findings confirmed that the proposed detection technique is able to detect obfuscated Tor Traffic efficiently and overcome the current research gaps. Due to the limitations of this study including interpretability and generalization to new obfuscation techniques; the following might be a future work: a potential future study on the detection of obfuscated Tor traffic using deep learning could involve the utilization of recurrent neural networks to examine network traffic patterns. ViT is particularly effective in analyzing sequential data, such as network traffic, and can be trained to recognize patterns that indicate obfuscated traffic. Additionally, incorporating feature engineering techniques, like packet header and payload analysis, can also enhance detection accuracy. Another avenue to explore could be the use of Re-

inforcement Learning algorithms to adapt to the dynamic changes in obfuscation techniques employed by attackers.

CRedit authorship contribution statement

The authors confirm contribution to the paper as follows:

Study conceptualization: Y. Sanjalawe and S. Al-E'mari;
 Methodology: Y. Sanjalawe, S. Al-E'mari, and S. Fraihat;
 Software: S. Al-E'mari;
 Validation: S. Fraihat;
 Formal Analysis: Y. Sanjalawe;
 Investigation: Y. Sanjalawe, S. Al-E'mari, and S. Fraihat;
 Resources: Y. Sanjalawe, S. Al-E'mari, and S. Fraihat;
 Data curation: S. Al-E'mari;
 Writing - original draft: Y. Sanjalawe, S. Al-E'mari;
 Writing - review and editing: Y. Sanjalawe and S. Fraihat;
 Visualization: Y. Sanjalawe, S. Al-E'mari;
 Supervision and administration: Y. Sanjalawe;
 Funding acquisition: S. Fraihat.

Declaration of competing interest

We confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Data availability

No data was used for the research described in the article.

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